

Functional Dependencies and Normalization

TDDD12 – Assignment 3

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Task 1

The following functional dependencies are given in the assignment.

FD1: $\{A\} \rightarrow \{B, C\}$

FD2: $\{C\} \rightarrow \{A, D\}$

FD3: $\{D, E\} \rightarrow \{F\}$

a)

The derivation of functional dependency ($\{C\} \rightarrow \{B\}$) using Armstrong rules.

FD4: $\{C\} \rightarrow \{A\}$ (Decomposition rule of FD2)

FD5: $\{A\} \rightarrow \{B\}$ (Decomposition rule of FD1)

FD6: $\{C\} \rightarrow \{B\}$ (Transitivity rule with FD4 and FD5)

b)

The derivation of functional dependency ($\{A, E\} \rightarrow \{F\}$) using Armstrong rules.

FD4: $\{A\} \rightarrow \{C\}$ (Decomposition rule with FD1)

FD5: $\{A\} \rightarrow \{A, D\}$ (Transitivity rule with FD4 and FD2)

FD6: $\{A\} \rightarrow \{D\}$ (Decomposition rule with FD5)

FD7: $\{A, E\} \rightarrow \{D, E\}$ (Augmentation of FD6 with E)

FD8: $\{A, E\} \rightarrow \{F\}$ (Augmentation of FD6 with E)

Task 2

R(A,B,C,D,E,F)

FD1: $\{A\} \rightarrow \{B, C\}$

FD2: $\{C\} \rightarrow \{A, D\}$

FD3: $\{D, E\} \rightarrow \{F\}$

a)

$X^+ = \{A, B, C, D\}$

b)

$X^+ = \{C, E, A, D, F, B\}$

Task 3

a)

The candidate keys for relation R are: $\{A,B\}$ and $\{A,D\}$.

b)

FD2 and FD3 violate the BCNF condition.

c)

First the relation R is decomposed into two separate relations, namely R1 and R2:

R2(E,F) with FD2, CK: $\{E\}$

R1(A,B,C,D,E) with FD4 and FD3

The functional dependency FD4 is derived from FD1 as follows:

FD4: $\{A,B\} \rightarrow \{C,D,E\}$ (Decomposition rule of FD1)

Now we check if R1 and R2 fulfills the BCNF condition. We can easily see that the only relation that does not fulfills BCNF condition is R1. Hence this relation should be decomposed further. The new relations are named R1A and R1B.

R1A(D,B) with FD3, CK: $\{D\}$

R1B(A,C,D,E) with FD7, CK: $\{A, D\}$

The functional dependency FD7 is derived as follows:

FD5: $\{A,D\} \rightarrow \{A,B\}$ (Augmentation rule of FD3)

FD6: $\{A,D\} \rightarrow \{C,D,E\}$ (Transitivity rule with FD4 and FD5)

FD7: $\{A,D\} \rightarrow \{C,E\}$ (Decomposition rule of FD6)

Finally the relation R(A, B, C, D, E, F) is decomposed to R2(E, F), R1A(D, B) and R1B(A, C, D, E), all of which are in normal form.

Task 4

The following information is given in the assignment.

$R(A,B,C,D,E)$
 $FD1: \{A,B,C\} \rightarrow \{D,E\}$
 $FD2: \{B,C,D\} \rightarrow \{A,E\}$
 $FD3: \{C\} \rightarrow \{D\}$

a)

FD3 violate the BCNF condition.

b)

First the relation R is decomposed into R1 and R2.

$R1(C,D)$ with FD3, CK: $\{C\}$
 $R2(A,B,C,E)$ with FD4 and FD6

where the functional dependencies FD4 and FD6 are derived below.

$FD4: \{A,B,C\} \rightarrow \{E\}$ (Decomposition rule of FD1)
 $FD5: \{B,C\} \rightarrow \{B,C,D\}$ (Augmentation rule of FD3 with $\{B,C\}$)
 $FD6: \{B,C\} \rightarrow \{A,E\}$ (Transitivity rule with FD5 and FD2)